Retrieval-Augmented Reward Modeling

Long-Horizon Robot Manipulation in Simulation
Proposal Overview + Weekly Execution Update

RAG-cross-RL

April 1, 2026

Outline

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Problem Setting

Why long-horizon manipulation is hard

- Sparse success rewards make exploration and credit assignment difficult.
- Hand-crafted shaping is labor-intensive and often breaks when tasks or layouts change.
- Intermediate states may look plausible while still being off-task.

What the project tries to fix

- Use retrieved evidence from demonstrations and stage descriptions.
- Turn that evidence into a denser notion of progress.
- Improve reward quality before worrying about policy quality.

Why Retrieval for Reward Modeling

Research context

- Embodied retrieval is already active [7, 5, 11].
- Retrieval for robotic demonstrations also exists [1, 3, 4].
- Reward modeling for robotics is growing quickly [8, 12, 9].

Scope of this proposal

- Simulation first, with controlled supervision and evaluation.
- No VLA pretraining or real-robot claim in the first stage.
- Focused question: can retrieval improve reward quality for long-horizon tasks?

Objectives and Research Questions

Objectives

- Build a retrieval-conditioned reward modeling pipeline in simulation.
- Validate a short-horizon prototype before full reward-head training.
- Measure whether retrieval improves reward estimation and downstream learning.

Research questions

- Does retrieved context improve progress estimation?
- Does retrieval-conditioned reward improve policy learning and OOD robustness?
- Which retrieval granularity works best: episode, stage, or sub-trajectory?

Hypotheses and Success Criteria

Working hypotheses

- H1: Multimodal retrieval helps distinguish real intermediate progress from misleading states.
- H2: Stage or sub-trajectory retrieval should beat full-episode retrieval because it reduces irrelevant context.
- H3: Retrieval helps more on long-horizon and OOD settings than on short in-distribution tasks.

What counts as success

- Better progress estimation than no-context and random-context baselines.
- A clean transition from retrieval to reward modeling and then to `IQL`.

Method Overview

Core formulation

$$r_t = f_\theta\big(o_t, g, \mathrm{Retrieve}(o_t, g, \mathcal{M})\big)$$

Memory bank

- Task instructions
- Stage descriptions
- Demonstration clips or sub-trajectories
- Progress labels and metadata

Retriever

- Encode current observation and task condition.
- Retrieve top-k stage-level evidence.
- Feed retrieved context into reward estimation.

Supervision and Training Path

Reward supervision

- Primary target: simulator-derived stage progress regression.
- Auxiliary target: pairwise ranking between earlier and later states.
- Labels come from simulator structure, not free-form human annotation.

Training plan

- Prototype first: retrieval-based progress proxy.
- Full model next: retrieval-conditioned reward head.
- Final integration: offline RL with `IQL`.

Benchmarks and OOD Protocol

Benchmarks

- Primary benchmark: `LIBERO` [6]
- Secondary option: `ManiSkill2/3` [2, 10]
- Core baselines: sparse reward, no context, random context, retrieved context

Protocol

- ID: seen tasks, objects, and layouts
- OOD-object: held-out object instances or assets
- OOD-layout: held-out scene configurations or poses

Ablations and Evaluation Metrics

Required ablations

- no context vs random context vs retrieved context
- text-only vs video-only vs multimodal retrieval
- episode vs stage vs sub-trajectory retrieval
- progress-only loss vs progress + ranking

Metrics

- Progress regression error and correlation
- Ranking accuracy
- Success rate and normalized return under `IQL`
- Robustness on OOD splits

Expected Contributions

What the proposal aims to contribute

- A clear formulation of retrieval-conditioned reward modeling.
- A practical supervision recipe derived from simulator signals.
- A protocol to evaluate reward quality before policy quality.
- A systematic comparison of retrieval granularity for long-horizon manipulation.

Execution Roadmap

Phase plan

- P0: one-task prototype with memory bank, retrieval, progress proxy, and visual artifacts
- P1: lock benchmark, OOD split, and supervision labels
- P2: build the full retrieval memory bank
- P3: train the retrieval-conditioned reward head
- P4: train offline RL with `IQL` using the learned reward
- P5: run ablations, analyze OOD robustness, and write up

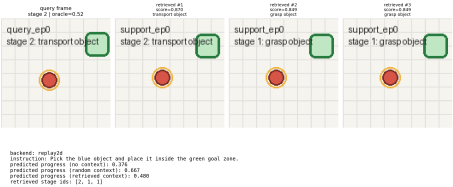
This Week's Progress: Implementation

Implemented this week

- Pinned the repo to `Python 3.8.20` with a single `uv` workflow.
- Added `demo/demo_v0.py` with `replay2d` and `LIBERO` backends.
- Built a mini memory bank, top-k retrieval, and a retrieval-weighted progress proxy.

Artifacts and validation

- Exported demo snapshot, retrieval examples, progress plot, and GIF.
- Confirmed a working end-to-end path from environment to report artifacts.



Demo snapshot with query, retrieved frames, and predicted progress

This Week's Progress: Metrics

Current evidence

- `replay2d`: 48 support, 24 query, retrieved-context MAE = 0.0797, corr = 0.9735
- `LIBERO`: 16 support, 8 query, retrieved-context MAE = 0.0973, corr = 0.9475

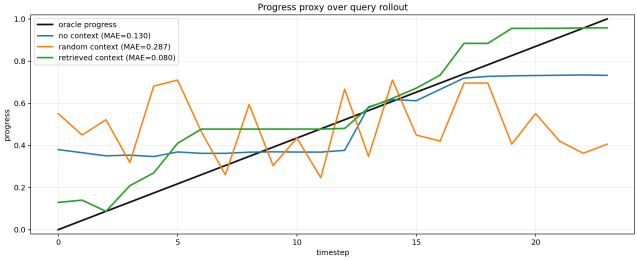
Interpretation

- Retrieved context beats both no-context and random-context baselines.
- The retrieval-to-progress path already works before full reward-head training.

Immediate next step

Replace the proxy with a retrieval-conditioned reward head, then connect it to offline RL with `IQL`.

This Week's Progress: Progress Plot



How to read it

- Oracle progress increases smoothly over time.
- Retrieved context tracks the oracle more closely than no context and random context.

Main takeaway

The prototype already shows that retrieval provides useful progress evidence before full reward-head training.

Takeaway

Main message

The proposal is not claiming a vague “robotic RAG” system. The target is narrower and testable: **use retrieval to improve reward quality for long-horizon manipulation in simulation, validate reward quality first, and only then move to policy learning with IQL.**

Status today

The project already has a working prototype that demonstrates the retrieval → progress estimation path. This reduces risk for the next phase, which is full reward-head training and downstream offline RL.

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